with are APIs that allow you to consume logs or metrics and provider console dashboards that allow viewing the same in a browser.

NO PATCHING

From an operational point of view, an important part of the maintenance you are outsourcing to your serverless compute provider is security patches, usually on the operating system and (language) runtime level. This means, while you still want to track vulnerabilities (especially on the application level), there is less work required on your end, besides testing.

AUTOSCALING

One prominent expectation in serverless compute is autoscaling. This means either vertical (using VMs with more or less CPU/RAM) or horizontal (adding or removing identical machines), changing the compute capacity in order to accommodate changing load. Serverless compute lets you handle autoscaling without having to worry about the details of the scaling models, which in turn reduces the number of metrics you need to ingest.

PAY AS YOU GO

Typically, being serverless comes with a pay-as-you-go model. In other words, if you do not use the capacity, you should, theoretically, not need to pay for it. There are exceptions to this (e.g., fixed fees for an orchestrator such as Kubernetes), and the billing granularity may be a factor as well. The implication here is that you need to make sure to track costs in an automated manner (including alerts), especially if you take the autoscaling into account.

From an observability point of view, serverless environments, in general, mean less effort is required, since a large part of the operational work is outsourced to the provider. A potential downside is the lack of visibility—if an OS or VM signal is not explicitly exposed, there is no way for you to access it.

CONTAINERS VS. FUNCTIONS

Serverless compute offerings come, roughly speaking, in two flavors: container centric and function as a service (FaaS). Let's take a quick look at what the major cloud providers offer in this space.

The container-centric serverless offering in Amazon Web Services (AWS) is AWS Fargate, which you can use as the data plane in the context of two container orchestrators, either in Amazon ECS on AWS Fargate (http://mng.bz/Rxrj) or Amazon EKS on AWS Fargate (http://mng.bz/2DR8). Depending on the orchestrator you're using, different signals are available. For example, EKS on Fargate offers a Fluent Bit–based log routing solution (http://mng.bz/D4Gy) as well as metrics (http://mng.bz/1qZ1) via CloudWatch, and ECS on Fargate offers monitoring (http://mng.bz/PzBY) via CloudWatch as well as metrics in Prometheus format.

AWS Lambda (http://mng.bz/JgWK) is the FaaS serverless compute offering. As depicted in figure 3.4, it provides a unified and comprehensive observability (http://mng.bz/j129) approach that includes AWS X-Ray (http://mng.bz/wv12). In this